

MATHEMATICS
Modern Algebra I
Suggested exercises 2

- (1) Let G be a group. For any $x, y \in G$, let $xyx^{-1}y^{-1}$ be the commutator of x, y . Let G^c be the subgroup generated by all commutators. Show that $G^c < G$ is a normal subgroup, G/G^c is abelian and if $H < G$ is any other normal subgroup such that G/H is abelian, then $G^c < H$.
- (2) Find all normal subgroups of S_3, S_4 .
- (3) Let $\text{Inn}(G)$ be the set of all inner automorphisms of G . Prove that $\text{Inn}(G)$ is a normal subgroup of G and $\text{Inn}(G) \cong G/Z(G)$.
- (4) Show that the group of non-zero real numbers with respect to multiplication, and the group of all real number under addition are not isomorphic.
- (5) Prove or disprove:
 - a) If K is a normal subgroup of H and H is a normal subgroup of G , then K is a normal subgroup of G .
 - b) If H is a normal subgroup of G and both $G, G/H$ are abelian, then G is abelian.
- (6) Let F be a field with q elements. What are the orders of $GL(n, F)$ and $SL(n, F)$.
- (7) Show that A_n is a normal subgroup of S_n .
- (8) Exercises 1, 2, 7, 9, 11 on page 75.

The midterms will be on Wednesday October 13 and Friday November 19.